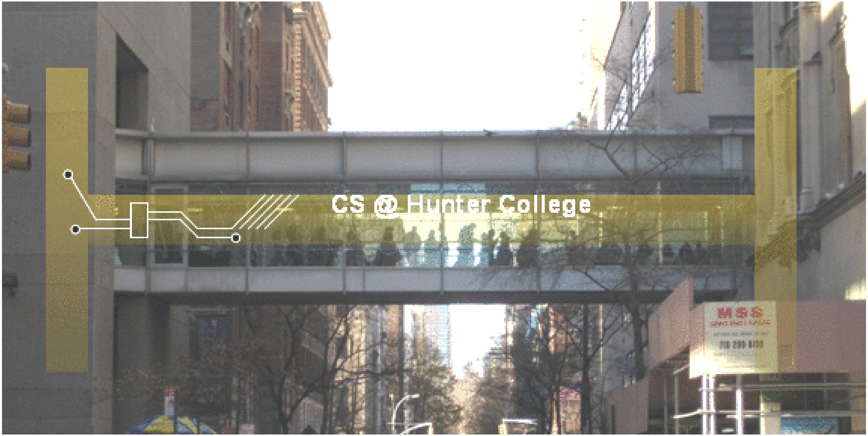
CSci 127: Introduction to Computer Science



hunter.cuny.edu/csci

Today's Topics

- Recap: Logical Expressions & Circuits
- Design: Cropping Images
- Accessing Formatted Data
- Design: Astrophysics and astropy

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- **Recap: Logical Expressions & Circuits**
- Design: Cropping Images
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Recap: Logical Operators

and

in1		in2	*returns:*
False	and	False	False
False	and	True	False
True	and	False	False
True	and	True	True

Recap: Logical Operators

and

in1		in2	*returns:*
False	and	False	False
False	and	True	False
True	and	False	False
True	and	True	True

or

in1		in2	*returns:*
False	or	False	False
False	or	True	True
True	or	False	True
True	or	True	True

Recap: Logical Operators

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in1		in2	*returns:*
False	and	False	False
False	and	True	False
True	and	False	False
True	and	True	True

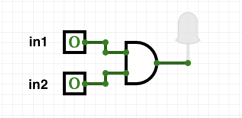
or

in1		in2	*returns:*
False	or	False	False
False	or	True	True
True	or	False	True
True	or	True	True

not

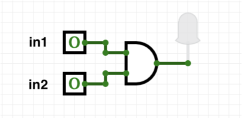
	in1	*returns:*
not	False	True
not	True	False

Logical Operators & Circuits



- Each logical operator (`and`, `or`, & `not`) can be used to join together expressions.

Logical Operators & Circuits



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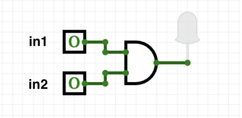
  Example: `in1 and in2`

Logical Operators & Circuits

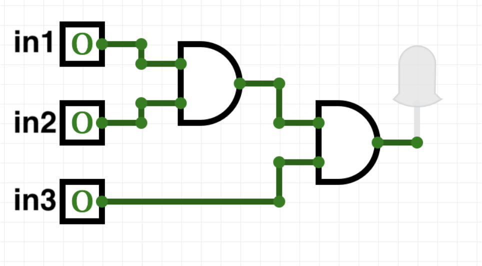


- Each logical operator (`and`, `or`, & `not`) can be used to join together expressions.

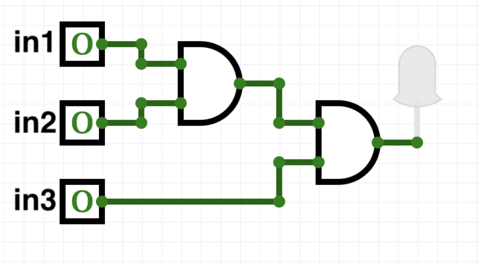
  Example: `in1 and in2`

- Each logical operator (`and`, `or`, & `not`) has a corresponding logical circuit that can be used to join together inputs.

Examples: Logical Circuit



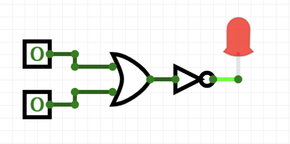
Examples: Logical Circuit

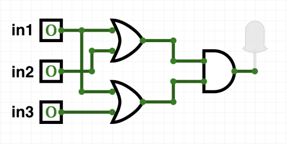


`(in1 and in2) and in3`

More Circuit Examples

Examples from last lecture:





Draw a circuit that corresponds to each logical expression:

- `not( in1 or in2 )`
- `(in1 or in2) and (in1 or in3)`
- `(not(in1 and not in2)) or (in1 and (in2 and in3))`

Challenge:

Predict what the code will do:

- ```
  x = 6
  y = x % 4
  w = y**3
  z = w // 2
  print(x,y,w,z)
  x,y = y,w
  print(x,y,w,z)
  x = y / 2
  print(x,y,w,z)
  ```
- ```
  sports = ["Field Hockey","Swimming","Water Polo"]
  mess = "Qoauxca BrletRce crcx qvBnqa ocUxk"
  result = ""
  for i in range(len(mess)):
      if i % 3 == 0:
          print(mess[i])
          result = result + mess[i]
  print(sports[1], result)
  ```

Python Tutor

```
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x,y = y,w
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(Demo with `pythonTutor`)

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- **Design: Cropping Images**
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Challenge: Design Question

From Final Exam, Fall 2017, V4, #6.

Design an algorithm that reads in an image and displays the lower left corner of the image.

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Design an algorithm that reads in an image and displays the lower left corner of the image.

Input:
Output:
Process: *(Brainstorm for a "To Do" list to accomplish this.)*

Design Question

Design a program that asks the user for an image and then display the upper left quarter of the image. (First, design the pseudocode, and if time, expand to a Python program.)

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  6. Display the new image.

In Pairs or Triples: Design Question

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6. Display the new image.
```
plt.imshow(img2) #Load our new image into pyplot
plt.show() #Show the image (waits until closed to continue)
```

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Structured Data

	Undergraduate		
College	Full-time	Part-time	Total
Baruch	11,288	3,922	15,210
Brooklyn	10,198	4,208	14,406
City	10,067	3,250	13,317
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John Jay	9,831	2,843	12,674
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Structured Data

pandas

$y_{it} = \beta' x_{it} + \mu_i + \epsilon_{it}$

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- To use, add to the top of your program:

```
import pandas as pd
```

CSV Files

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- Columns are separated by commas on each line.

CSV Files

```
Source:  https://en.wikipedia.org/wiki/Demographics_of_New_York_City,,,,,,
All population figures are consistent with present-day boundaries.,,,,,,
First census after the consolidation of the five boroughs,,,,,,
,,,,,,
,,,,,,
Year,Manhattan,Brooklyn,Queens,Bronx,Staten Island,Total
1698,4937,2017,,,727,7681
1771,21863,3623,,,2847,28423
1790,33131,4549,6159,1781,3827,49447
1800,60515,5740,6642,1755,4563,79215
1810,96373,8303,7444,2267,5347,119734
1820,123706,11187,8246,2782,6135,152056
1830,202589,20535,9049,3023,7082,242278
1840,312710,47613,14480,5346,10965,391114
1850,515547,138882,18593,8032,15061,696115
1860,813669,279122,32903,23593,25492,1174779
1870,942292,419921,45468,37393,33029,1478103
1880,1164673,599495,56559,51980,38991,1911698
1890,1441216,838547,87050,88908,51693,2507414
1900,1850093,1166582,152999,200507,67021,3437202
1910,2331542,1634351,284041,430980,85969,4766883
1920,2284103,2018356,469042,732016,116531,5620048
1930,1867312,2560401,1079129,1265258,158346,6930446
1940,1889924,2698285,1297634,1394711,174441,7454995
1950,1960101,2738175,1550849,1451277,191555,7891957
1960,1698281,2627319,1809578,1424815,221991,7781984
1970,1539233,2602012,1986473,1471701,295443,7894862
1980,1428285,2230936,1891325,1168972,352121,7071639
1990,1487536,2300664,1951598,1203789,378977,7322564
2000,1537195,2465326,2229379,1332650,443728,8008278
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2015,1644518,2636735,2339150,1455444,474558,8550405
```

`nycHistPop.csv`

Reading in CSV Files

	Undergraduate		
College	Full-time	Part-time	Total
Baruch	11,288	3,922	15,210
Brooklyn	10,198	4,208	14,406
City	10,067	3,250	13,317
Hunter	12,223	4,500	16,723
John Jay	9,831	2,843	12,674
Lehman	6,600	4,720	11,320
Medgar Evers	4,760	2,059	6,819
NYCCT	10,912	6,370	17,282
Queens	11,693	4,633	16,326
Staten Island	9,584	2,948	12,532
York	5,066	3,192	8,258

- To read in a CSV file: `myVar = pd.read_csv("myFile.csv")`

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- Often abbreviated: `df`.
- It also has **Series**, that is perfect for holding a row or column of data.

Example: Reading in CSV Files

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Source:  https://en.wikipedia.org/wiki/Demographics_of_New_York_City,,,,,,
All population figures are consistent with present-day boundaries.,,,,,,
First census after the consolidation of the five boroughs,,,,,,
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1698,4937,2017,,,727,7681
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1860,813669,279122,32903,23593,25492,1174779
1870,942292,419921,45468,37393,33029,1478103
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1940,1889924,2698285,1297634,1394711,174441,7454995
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`nycHistPop.csv`

In Lab 6

Example: Reading in CSV Files

```
import matplotlib.pyplot as plt
import pandas as pd
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2015,1644518,2636735,2339150,1455444,474558,8550405
```

`nycHistPop.csv`

In Lab 6

Example: Reading in CSV Files

```python
import matplotlib.pyplot as plt
import pandas as pd

pop = pd.read_csv('nycHistPop.csv',skiprows=5)
```

```
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nycHistPop.csv

In Lab 6

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nycHistPop.csv

In Lab 6

```python
import matplotlib.pyplot as plt
import pandas as pd

pop = pd.read_csv('nycHistPop.csv',skiprows=5)

pop.plot(x="Year")
plt.show()
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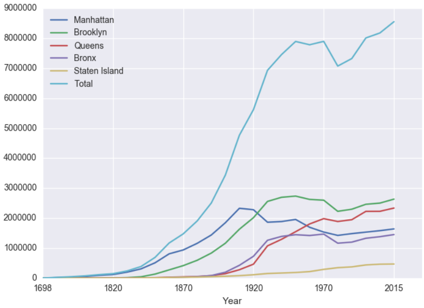
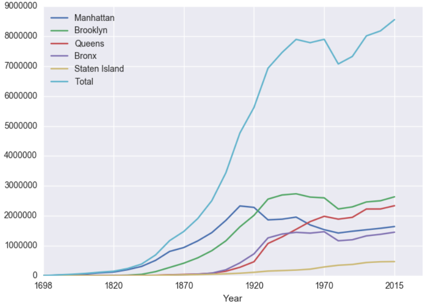
nycHistPop.csv

In Lab 6

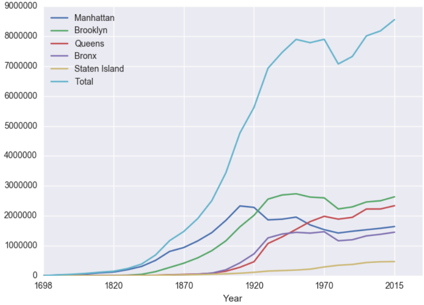


Series in Pandas



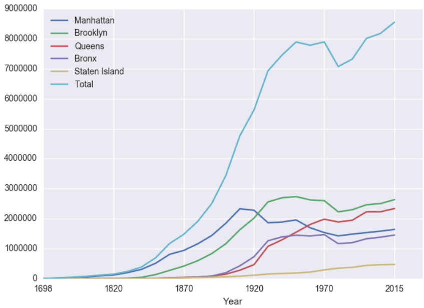
- Series can store a column or row of a DataFrame.

Series in Pandas



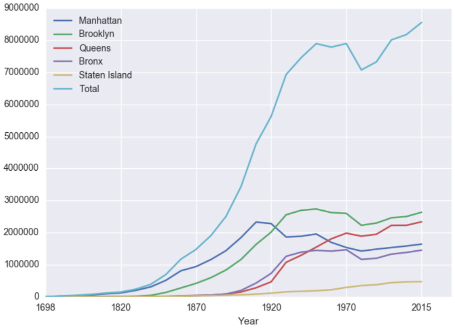
- Series can store a column or row of a DataFrame.
- Example: `pop["Manhattan"]` is the Series corresponding to the column of Manhattan data.

Series in Pandas



- Series can store a column or row of a DataFrame.
- Example: `pop["Manhattan"]` is the Series corresponding to the column of Manhattan data.
- Example:
  ```
  print("The largest number living in the Bronx is",
  pop["Bronx"].max())
  ```

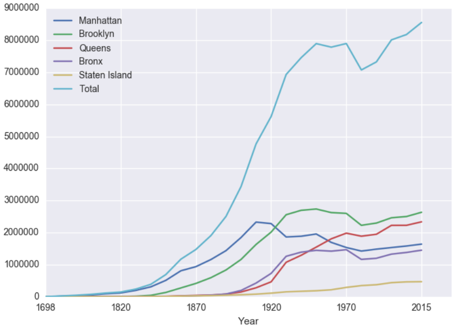
Challenge:



Predict what the following will do:

- `print("Queens:", pop["Queens"].min())`

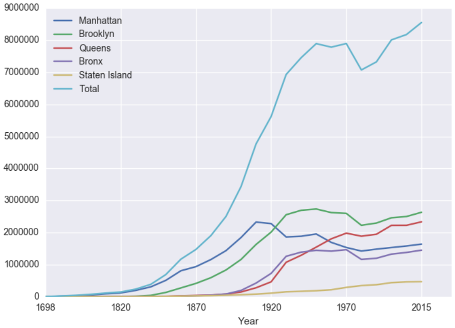
Challenge:



Predict what the following will do:

- `print("Queens:", pop["Queens"].min())`
- `print("S I:", pop["Staten Island"].mean())`

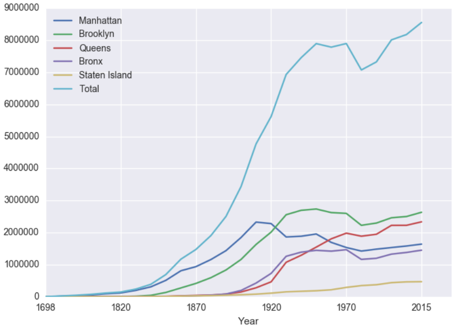
Challenge:



Predict what the following will do:

- `print("Queens:", pop["Queens"].min())`
- `print("S I:", pop["Staten Island"].mean())`
- `print("S I:", pop["Staten Island"].std())`

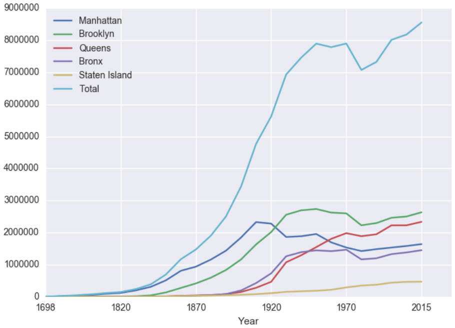
Challenge:



Predict what the following will do:

- `print("Queens:", pop["Queens"].min())`
- `print("S I:", pop["Staten Island"].mean())`
- `print("S I:", pop["Staten Island"].std())`
- `pop.plot.bar(x="Year")`

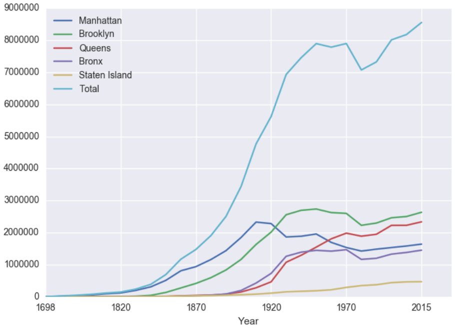
Challenge:



Predict what the following will do:

- `print("Queens:", pop["Queens"].min())`
- `print("S I:", pop["Staten Island"].mean())`
- `print("S I:", pop["Staten Island"].std())`
- `pop.plot.bar(x="Year")`
- `pop.plot.scatter(x="Brooklyn", y= "Total")`

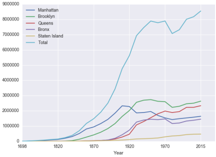
Challenge:



Predict what the following will do:

- `print("Queens:", pop["Queens"].min())`
- `print("S I:", pop["Staten Island"].mean())`
- `print("S I:", pop["Staten Island"].std())`
- `pop.plot.bar(x="Year")`
- `pop.plot.scatter(x="Brooklyn", y= "Total")`
- `pop["Fraction"] = pop["Bronx"]/pop["Total"]`

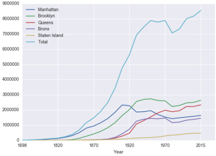
Solutions



Predict what the following will do:

- `print("Queens:", pop["Queens"].min())`

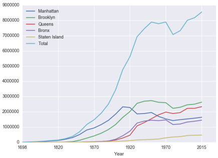
Solutions



Predict what the following will do:

- `print("Queens:", pop["Queens"].min())`
  Minimum value in the column with label "Queens".

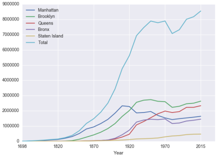
Solutions



Predict what the following will do:

- `print("Queens:", pop["Queens"].min())`
  Minimum value in the column with label "Queens".
- `print("S I:", pop["Staten Island"].mean())`

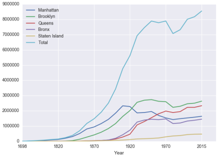
Solutions



Predict what the following will do:

- `print("Queens:", pop["Queens"].min())`
  Minimum value in the column with label "Queens".
- `print("S I:", pop["Staten Island"].mean())`
  Average of values in the column "Staten Island".

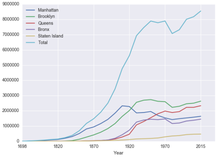
Solutions



Predict what the following will do:

- `print("Queens:", pop["Queens"].min())`
  Minimum value in the column with label "Queens".
- `print("S I:", pop["Staten Island"].mean())`
  Average of values in the column "Staten Island".
- `print("S I :", pop["Staten Island"].std())`

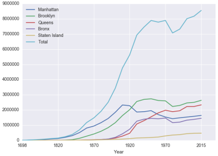
Solutions



Predict what the following will do:

- `print("Queens:", pop["Queens"].min())`
  Minimum value in the column with label "Queens".
- `print("S I:", pop["Staten Island"].mean())`
  Average of values in the column "Staten Island".
- `print("S I :", pop["Staten Island"].std())`
  Standard deviation of values in the column "Staten Island".

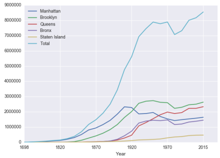
Solutions



Predict what the following will do:

- `print("Queens:", pop["Queens"].min())`
  Minimum value in the column with label "Queens".
- `print("S I:", pop["Staten Island"].mean())`
  Average of values in the column "Staten Island".
- `print("S I :", pop["Staten Island"].std())`
  Standard deviation of values in the column "Staten Island".
- `pop.plot.bar(x="Year")`

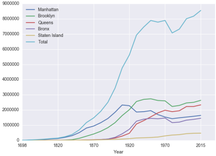
Solutions



Predict what the following will do:

- `print("Queens:", pop["Queens"].min())`
  Minimum value in the column with label "Queens".
- `print("S I:", pop["Staten Island"].mean())`
  Average of values in the column "Staten Island".
- `print("S I :", pop["Staten Island"].std())`
  Standard deviation of values in the column "Staten Island".
- `pop.plot.bar(x="Year")`
  Bar chart with x-axis "Year".

Solutions



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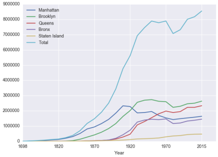
  Minimum value in the column with label "Queens".
- `print("S I:", pop["Staten Island"].mean())`
  Average of values in the column "Staten Island".
- `print("S I :", pop["Staten Island"].std())`
  Standard deviation of values in the column "Staten Island".
- `pop.plot.bar(x="Year")`
  Bar chart with x-axis "Year".
- `pop.plot.scatter(x="Brooklyn", y= "Total")`

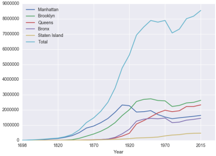
Solutions



Predict what the following will do:

- `print("Queens:", pop["Queens"].min())`
  Minimum value in the column with label "Queens".
- `print("S I:", pop["Staten Island"].mean())`
  Average of values in the column "Staten Island".
- `print("S I :", pop["Staten Island"].std())`
  Standard deviation of values in the column "Staten Island".
- `pop.plot.bar(x="Year")`
  Bar chart with x-axis "Year".
- `pop.plot.scatter(x="Brooklyn", y= "Total")`
  Scatter plot of Brooklyn versus Total values.

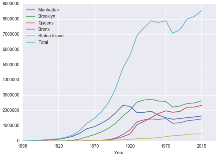
Solutions



Predict what the following will do:

- `print("Queens:", pop["Queens"].min())`
  Minimum value in the column with label "Queens".
- `print("S I:", pop["Staten Island"].mean())`
  Average of values in the column "Staten Island".
- `print("S I :", pop["Staten Island"].std())`
  Standard deviation of values in the column "Staten Island".
- `pop.plot.bar(x="Year")`
  Bar chart with x-axis "Year".
- `pop.plot.scatter(x="Brooklyn", y= "Total")`
  Scatter plot of Brooklyn versus Total values.
- `pop["Fraction"] = pop["Bronx"]/pop["Total"]`

Solutions



Predict what the following will do:

- `print("Queens:", pop["Queens"].min())`
  Minimum value in the column with label "Queens".
- `print("S I:", pop["Staten Island"].mean())`
  Average of values in the column "Staten Island".
- `print("S I :", pop["Staten Island"].std())`
  Standard deviation of values in the column "Staten Island".
- `pop.plot.bar(x="Year")`
  Bar chart with x-axis "Year".
- `pop.plot.scatter(x="Brooklyn", y= "Total")`
  Scatter plot of Brooklyn versus Total values.
- `pop["Fraction"] = pop["Bronx"]/pop["Total"]`
  New column with the fraction of population that lives in the Bronx.

Challenge:

	Undergraduate		
College	Full-time	Part-time	Total
Baruch	11,288	3,922	15,210
Brooklyn	10,198	4,208	14,406
City	10,067	3,250	13,317
Hunter	12,223	4,500	16,723
John Jay	9,831	2,843	12,674
Lehman	6,600	4,720	11,320
Medgar Evers	4,760	2,059	6,819
NYCCT	10,912	6,370	17,282
Queens	11,693	4,633	16,326
Staten Island	9,584	2,948	12,532
York	5,066	3,192	8,258

`cunyF2016.csv`

Write a complete Python program that reads in the file, `cunyF2016.csv`, and produces a scatter plot of full-time versus part-time enrollment.

Challenge:

Undergraduate			
College	Full-time	Part-time	Total
Baruch	11,288	3,922	15,210
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Solution:

1. *Include pandas & pyplot libraries.*

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Write a complete Python program that reads in the file, `cunyF2016.csv`, and produces a scatter plot of full-time versus part-time enrollment.

Solution:

1. *Include pandas & pyplot libraries.*
2. *Read in the CSV file.*
3. *Set up a scatter plot.*
4. *Display plot.*

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```
2. *Read in the CSV file.*
```
pop=pd.read_csv('cunyF2016.csv',skiprows=1)
```

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```
2. *Read in the CSV file.*
```
pop=pd.read_csv('cunyF2016.csv',skiprows=1)
```
3. *Set up a scatter plot.*
```
pop.plot.scatter(x="Full-
time",y="Part-time")
```

Challenge:

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3. *Set up a scatter plot.*
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time",y="Part-time")
```
4. *Display plot.*

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Write a complete Python program that reads in the file, `cunyF2016.csv`, and produces a scatter plot of full-time versus part-time enrollment.

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import pandas as pd
```
2. *Read in the CSV file.*
```
pop=pd.read_csv('cunyF2016.csv',skiprows=1)
```
3. *Set up a scatter plot.*
```
pop.plot.scatter(x="Full-
time",y="Part-time")
```
4. *Display plot.*
```
plt.show()
```

groupby()

Sometimes you have **recurring values in a column** and you want to examine the data for a particular value.

Rain in Australia				
Date	Location	MinTemp	MaxTemp	Rainfall
12/1/08	Albury	13.4	22.9	0.6
5/22/15	BadgerysCree	11	15.6	1.6
3/17/11	BadgerysCree	18.1	25.8	16.6
7/27/10	Cobar	5.3	17.2	0
9/5/10	Moree	12.1	19.8	23.4
1/23/12	CoffsHarbour	20	24.4	28
7/15/11	Moree	2.8	19	0
1/28/10	Newcastle	22.2	28	0
12/2/15	Moree	20.1	32	4.8
	...			

`AustraliaRain.csv`

groupby()

Sometimes you have **recurring values in a column** and you want to examine the data for a particular value.

For example, to **find the average rainfall at each location***:*

Rain in Australia				
Date	**Location**	**MinTemp**	**MaxTemp**	**Rainfall**
12/1/08	Albury	13.4	22.9	0.6
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7/15/11	Moree	2.8	19	0
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	...			

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Sometimes you have **recurring values in a column** and you want to examine the data for a particular value.

For example, to **find the average rainfall at each location***:*

1. *Import libraries.*

```
import pandas as pd
```

groupby()

Rain in Australia

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For example, to **find the average rainfall at each location***:*

1. *Import libraries.*
   ```
   import pandas as pd
   ```
2. *Read in the CSV file.*
   ```
   rain =
   pd.read_csv('AustraliaRain.csv',skiprows=1)
   ```

groupby()

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	...			

`AustraliaRain.csv`

For example, to **find the average rainfall at each location***:*

1. *Import libraries.*
   `import pandas as pd`
2. *Read in the CSV file.*
   `rain =`
   `pd.read_csv('AustraliaRain.csv',skiprows=1)`
3. *Group the data by location.*
   `groupAvg = rain.`**groupby('Location')**

groupby()

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Date	Location	MinTemp	MaxTemp	Rainfall
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	...			

`AustraliaRain.csv`

Sometimes you have **recurring values in a column** and you want to examine the data for a particular value.

For example, to **find the average rainfall at each location***:*

1. *Import libraries.*
   `import pandas as pd`
2. *Read in the CSV file.*
   `rain = pd.read_csv('AustraliaRain.csv',skiprows=1)`
3. *Group the data by location.*
   `groupAvg = rain.`**groupby('Location')**
4. *Print the average rainfall at each location.*
   `print(groupAvg['Rainfall'].mean())`

groupby()

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12/2/15	Moree	20.1	32	4.8
	. . .			

`AustraliaRain.csv`

```
Adelaide         1.572185
Albany           2.255073
Albury           1.925710
AliceSprings     0.869355
BadgerysCreek    2.207925
Ballarat         1.688830
Bendigo          1.621452
Brisbane         3.160536
Cairns           5.765317
Canberra         1.735038
Cobar            1.129262
CoffsHarbour     5.054592
Dartmoor         2.148554
```

Sometimes you have **recurring values in a column** and you want to examine the data for a particular value.

For example, to **find the average rainfall at each location***:*

1. *Import libraries.*
   `import pandas as pd`
2. *Read in the CSV file.*
   `rain = pd.read_csv('AustraliaRain.csv',skiprows=1)`
3. *Group the data by location.*
   `groupAvg = rain.`**groupby('Location')**
4. *Print the average rainfall at each location.*
   `print(groupAvg['Rainfall'].mean())`

groupby()

Sometimes you have **recurring values in a column** and you want to examine the data for a particular value.

For example, to **find the average rainfall at one location, e.g. Albury***:*

Rain in Australia

Date	Location	MinTemp	MaxTemp	Rainfall
12/1/08	Albury	13.4	22.9	0.6
5/22/15	BadgerysCree	11	15.6	1.6
3/17/11	BadgerysCree	18.1	25.8	16.6
7/27/10	Cobar	5.3	17.2	0
9/5/10	Moree	12.1	19.8	23.4
1/23/12	CoffsHarbour	20	24.4	28
7/15/11	Moree	2.8	19	0
1/28/10	Newcastle	22.2	28	0
12/2/15	Moree	20.1	32	4.8
	...			

`AustraliaRain.csv`

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`AustraliaRain.csv`

1. *Import libraries.*
   ```
   import pandas as pd
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2. *Read in the CSV file.*
   ```
   rain =
   pd.read_csv('AustraliaRain.csv',skiprows=1)
   ```
3. **Group the data by location get data for group Albury.**
   ```
   AlburyAvg =
   rain.groupby('Location').get_group('Albury')
   ```

groupby()

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`AustraliaRain.csv`

```
1.9257104647275156
```

Design Challenge

Stars						
Temperature (K)	**Luminosity(L/Lo)**	**Radius(R/Ro)**	**Absolute magnitude(Mv)**	**Star type**	**Star color**	**Spectral Class**
3068	0.0024	0.17	16.12	Brown Dwarf	Red	M
25000	0.056	0.0084	10.58	White Dwarf	Blue White	B
2650	0.00069	0.11	17.45	Brown Dwarf	Red	M
11790	0.00015	0.011	12.59	White Dwarf	Yellowish White	F
15276	1136	7.2	-1.97	Main Sequence	Blue-white	B
5800	0.81	0.9	5.05	Main Sequence	yellow-white	F
16500	0.013	0.014	11.89	White Dwarf	Blue White	B
3192	0.00362	0.1967	13.53	Red Dwarf	Red	M
6380	1.35	0.98	2.93	Main Sequence	yellow-white	F
3834	272000	1183	-9.2	Hypergiant	Red	M

- Design an algorithm that:
  - Prints the luminosity of the brightest star.
  - Prints the temperature of the coldest star.
  - Prints the average radius of a Hypergiant.

Design Challenge - Solution

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Temperature (K)	Luminosity(L/Lo)	Radius(R/Ro)	Absolute magnitude(Mv)	Star type	Star color	Spectral Class
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- **Libraries:** pandas

Design Challenge - Solution

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- **Libraries:** pandas
- **Process:**
  - Print **max** of **'Luminosity'** column

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- **Libraries:** pandas
- **Process:**
    - Print **max** of **'Luminosity'** column
    - Print **min** of **'Temperature'** column

Design Challenge - Solution

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- **Libraries:** pandas
- **Process:**
  - Print **max** of **'Luminosity'** column
  - Print **min** of **'Temperature'** column
  - **groupby 'Star Type'** and take **averages**, then print **max of 'Radius'** column

Design Challenge - Solution

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- **Libraries:** pandas
- **Process:**
  - Print **max** of **'Luminosity'** column
  - Print **min** of **'Temperature'** column
  - **groupby 'Star Type'** and take **averages**, then print **max of 'Radius'** column
  - OR **groupby 'Star Type'** and **get group 'Hypergiant'** to print **average 'Radius'**

Design Challenge - Code

- **Libraries:** pandas

```
import pandas as pd
stars = pd.read_csv('Stars.csv')
```

Design Challenge - Code

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  ```
  import pandas as pd
  stars = pd.read_csv('Stars.csv')
  ```

- **Process:**
  - Print **max** of **'Luminosity'** column

    ```
    print(stars['Luminosity(L/Lo)'].max())
    ```

Design Challenge - Code

- **Libraries:** pandas
  ```
  import pandas as pd
  stars = pd.read_csv('Stars.csv')
  ```
- **Process:**
  - Print **max** of **'Luminosity'** column
    ```
    print(stars['Luminosity(L/Lo)'].max())
    ```
  - Prints **min** of **'Temperature'** column and store it in temp variable
    ```
    print( stars['Temperature (K)'].min())
    ```

Design Challenge - Code

- **Libraries:** pandas

  ```
  import pandas as pd
  stars = pd.read_csv('Stars.csv')
  ```

- **Process:**
  - Print **max** of **'Luminosity'** column

    ```
    print(stars['Luminosity(L/Lo)'].max())
    ```

  - Prints **min** of **'Temperature'** column and store it in temp variable

    ```
    print( stars['Temperature (K)'].min())
    ```

  - **groupby 'Star Type'** and take **averages**, then print **max of 'Radius'** column

    ```
    print(stars.groupby('Star type')\
    .mean()['Radius(R/Ro)'].max())
    ```

Design Challenge - Code

- **Libraries:** pandas

  ```
  import pandas as pd
  stars = pd.read_csv('Stars.csv')
  ```

- **Process:**
  - Print **max** of **'Luminosity'** column

    ```
    print(stars['Luminosity(L/Lo)'].max())
    ```

  - Prints **min** of **'Temperature'** column and store it in temp variable

    ```
    print( stars['Temperature (K)'].min())
    ```

  - OR **groupby 'Star Type'** and **get group 'Hypergiant'** to print **average 'Radius'**

    ```
    print(stars.groupby('Star type')\
    .get_group('Hypergiant').mean()['Radius(R/Ro)'])
    ```

Recap

- Recap: Logical Expressions & Circuits

Recap

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- Accessing Formatted Data:
  - Pandas library has elegant solutions for accessing & analyzing structured data.

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- Accessing Formatted Data:
  - Pandas library has elegant solutions for accessing & analyzing structured data.
  - Can manipulate individual columns or rows ('Series').

Recap

- Recap: Logical Expressions & Circuits
- Accessing Formatted Data:
    - Pandas library has elegant solutions for accessing & analyzing structured data.
    - Can manipulate individual columns or rows ('Series').
    - Has useful functions for the entire sheet ('DataFrame') such as plotting.